CR-001: Advanced Medicine Search and Filtering

1. Change Request Details

Field	Details
Change Request ID	CR-001
Title	Advanced Medicine Search and Filtering
Priority	Medium
Difficulty	Easy–Medium
Change Type	Frontend Enhancement
Affected Version	Pharmacy Management System v1.0 baseline (assumed)
Dependency	Existing medicine list, category and supplier data

2. Business Requirement

Pharmacy staff need to locate medicines quickly when reviewing inventory. Basic search by medicine name, batch number, or category does not provide a combined view of supplier, price, and stock availability.

The system should support multiple simultaneous filters so authorized users can narrow the medicine list without manually inspecting every record.

3. Current Behaviour

- The medicine management page provides basic medicine search and listing.
- Users cannot combine supplier, price range, and stock-status criteria in a single filtering workflow.
- Existing medicine add, edit, delete, and stock-related actions must remain available.

4. Requested Change

Add an advanced filtering panel to the Medicine Management page. The implementation must:

- Search medicine names using partial, case-insensitive matching.
- Filter by medicine category and supplier.
- Filter by inclusive minimum and maximum selling price.
- Filter by stock status: Available, Low Stock, or Out of Stock.
- Apply all active filters together (AND logic).
- Provide Clear All Filters and show the matching medicine count.
- Show a useful no-results message; preserve correct medicine identifiers and existing actions.
- Validate numeric price inputs and handle missing optional values safely.

5. Scope

In Scope

- Medicine search, category and supplier selectors, price inputs, stock-status selector.

- Combined filtering, matching count, clear button, empty state, responsive styling.
- Regression checks for existing medicine management functions.

Out of Scope

- Database schema changes, new medicine fields, supplier creation, purchasing or reorder workflows.
- New API endpoints and server-side filtering unless existing medicine data is insufficient.
- Changes to authentication and authorization.

6. Expected / Potentially Affected Files

- public/pages/medicines.html
- public/js/medicines.js
- public/css/medicine.css (verify actual stylesheet name)
- routes/medicines.js (only if existing response does not supply required fields)

Any additional files changed must be identified and justified in the implementation report.

7. Functional & Security Requirements

- FR-001: Medicine-name search shall support partial, case-insensitive matching.
- FR-002: Category filtering shall display medicines in the selected category.
- FR-003: Supplier filtering shall display medicines from the selected supplier.
- FR-004: Minimum and maximum price shall be applied inclusively; invalid ranges shall be handled clearly.
- FR-005: Stock status shall use existing inventory/reorder-level rules; zero stock shall be Out of Stock.
- FR-006: Multiple active filters shall use AND logic.
- FR-007: Clear All Filters shall restore the unfiltered medicine list.
- FR-008: Matching count and no-results message shall reflect visible records.
- FR-009: Existing medicine CRUD actions shall continue to target the correct medicine ID.
- SEC-001: Existing authentication and role permissions shall remain unchanged.
- SEC-002: Filtering shall not expose unauthorized data or render user input as unsafe HTML.

8. Acceptance Criteria

- AC-001: Partial and mixed-case medicine-name searches return correct matches.
- AC-002: Category and supplier filters each return correct records.
- AC-003: Minimum/maximum price boundaries are inclusive and invalid values are handled.
- AC-004: Available, Low Stock, and Out of Stock follow documented stock rules.
- AC-005: Combining filters returns only medicines satisfying all active criteria.
- AC-006: Result count equals the number of visible medicines.
- AC-007: No-match searches show a clear empty state.
- AC-008: Clear All Filters restores the complete permitted list.
- AC-009: Existing medicine edit/delete and authorization still work correctly.

9. Expected Outcome

Authorized pharmacy users can find relevant medicines quickly using multiple criteria, with clear result feedback and no loss of existing medicine management functionality.

10. Implementation Report Requirement

- Implementation summary and actual affected files.
- Filtering rules, stock-status definition, and technical/security considerations.
- Test cases, observed results, acceptance-criteria verification, and screenshots.
- Final implementation status and any outstanding issues.